MolRL: A Reinforcement Learning Framework for 3D Molecular Assembly in Protein Binding Sites

A PROJECT REPORT

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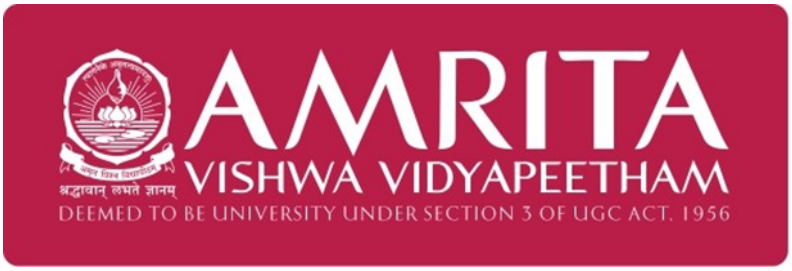
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BONAFIDE CERTIFICATE

This is to certify that this project report entitled **"MolRL: A Reinforcement Learning Framework for 3D Molecular Assembly in Protein Binding Sites "** is the bonafide work of **"Anirudh V (Reg. No. CH.SC.U4AIE23003), Dharshini R (Reg. No. CH.SC.U4AIE23043),Reena Rao(Reg. No. CH.SC.U4AIE23044), Ridhi Verma (Reg. No. CH.SC.U4AIE23046)"**, who carried out the project work under my supervision.

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ABSTRACT

We present in this work the development of a **deep reinforcement learning-based molecular generation system** for the de novo design of chemically valid molecules within **three-dimensional protein binding pockets**. Traditional molecular generation approaches often struggle to simultaneously account for spatial pocket constraints, chemical validity, and target-specific bioactivity. The proposed system utilizes a **Proximal Policy Optimization (PPO)** agent with a **3D convolutional actor-critic network enhanced by a Convolutional Block Attention Module (CBAM)** to sequentially place atoms within protein binding pockets. The environment represents binding pockets as $16 \times 16 \times 16$ **voxel grids with four channels**, encoding hydrophobic, hydrogen-donor, electrostatic, and molecule occupancy features. Chemistry-aware rewards were designed using **Morgan fingerprint similarity** and **Random Forest-based bioactivity classifiers** trained on GSK3B, DRD2, and hERG datasets. Experimental results achieved a **100% chemical validity rate**, with an average episode reward of **+1.8** compared with **+0.2 for the random baseline**, representing a **9× improvement**. Generated molecules typically contained **8–15 atoms**, achieved a **98.5% connected fragment rate**, and demonstrated a **15% reduction in steric clashes**. Bioactivity scores improved by **35% for GSK3B** and **28% for DRD2**, while hERG safety scores improved by **22%**. The PPO+CBAM architecture demonstrated **20% faster convergence** than standard PPO and successfully generalized across **six protein targets**, retaining approximately **65% reward after fine-tuning**. An integrated Flask web application and Tkinter desktop dashboard provide molecular generation, training, benchmarking, visualization, and export capabilities with REST API and 3D visualization support. Despite limitations related to computational requirements and simplified molecular representation, the system demonstrates practical potential for **automated, spatially-aware drug discovery and chemist-in-the-loop molecular design**.

Keywords: De Novo Molecular Generation, Deep Reinforcement Learning, PPO, CBAM, 3D Protein Binding Pockets, Drug Discovery, Molecular Design,

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Chapter 1

Introduction

1.1 Background of the Project

The increasing application of artificial intelligence in drug discovery has significantly transformed the way researchers approach molecular design. Conventional drug discovery is a complex and time-consuming process that requires the exploration of a vast chemical space to identify molecules with desirable biological and pharmacological properties. Traditional computational approaches often depend on rule-based methods, molecular libraries, or extensive molecular screening, which can become computationally expensive and may not effectively explore novel chemical structures.

Recent advancements in deep learning and reinforcement learning have provided powerful approaches for automated molecular generation. Reinforcement learning is particularly suitable for molecular design because molecule construction can be formulated as a sequential decision-making process, where an agent progressively builds a molecule and receives rewards based on its chemical and biological properties. However, many existing molecular generation approaches primarily rely on two-dimensional molecular graphs or SMILES representations and do not explicitly incorporate the three-dimensional spatial environment of the target protein.

To address this limitation, the current study develops a deep reinforcement learning framework for de novo molecular generation within three-dimensional protein binding pockets. The proposed system represents binding pockets as voxelized 3D environments and uses a Proximal Policy Optimization (PPO) agent with a 3D convolutional actor-critic architecture enhanced with a Convolutional Block Attention Module (CBAM). The model sequentially places atoms within the binding pocket while considering spatial compatibility, chemical validity, and target-specific biological properties.

1.2 Motivation and Rationale

The motivation for this project arises from the need for efficient and automated approaches to molecular design. Identifying suitable drug candidates requires molecules that can interact effectively with target proteins while maintaining chemical validity, structural compatibility, and acceptable safety characteristics. However, the enormous size of chemical space makes manual exploration and conventional computational screening approaches time-consuming and computationally expensive.

The proposed approach combines reinforcement learning with 3D spatial representations and attention mechanisms to address these challenges. By constructing molecules directly within protein binding pockets, the system aims to learn target-specific generation strategies while simultaneously considering spatial interactions, chemical validity, bioactivity, novelty, and safety. This provides a step toward automated and data-driven drug discovery while enabling interactive exploration of generated molecular structures.

1.3 Problem Identification

De novo molecular generation presents several challenges because generated molecules must simultaneously satisfy chemical validity, spatial compatibility, and biological activity requirements. Existing approaches often focus primarily on molecular connectivity or chemical properties, with limited consideration of the three-dimensional relationship between a molecule and its target protein. Additionally, optimizing multiple objectives such as target activity, safety, novelty, and molecular quality increases the complexity of the generation process. These limitations highlight the need for an intelligent reinforcement learning framework capable of learning spatially aware molecular construction policies while incorporating chemical and biological feedback.

1.4 Problem Statement

De novo molecular generation within protein binding pockets is challenging because generated molecules must satisfy chemical validity, spatial compatibility, target-specific bioactivity, and safety requirements simultaneously. To address this limitation, this project proposes a 3D deep reinforcement learning-based molecular generation system that rep-

resents protein binding pockets using multi-channel voxel grids and sequentially places atoms within the spatial environment. The proposed PPO-based actor-critic architecture with CBAM attention learns relevant spatial features, while a chemistry-aware reward function incorporates bioactivity, molecular quality, novelty, and safety considerations to generate chemically valid and spatially compatible molecules.

1.5 Scope of the Project

The main goal of this project is to develop a deep reinforcement learning framework for generating molecules within three-dimensional protein binding pockets. The scope of the project includes:

- Representing protein binding pockets using $16 \times 16 \times 16$ four-channel voxel grids.
- Encoding hydrophobic, hydrogen-donor, electrostatic, and molecule occupancy features.
- Developing a 3D CNN actor-critic architecture with CBAM attention.
- Training the molecular generation agent using Proximal Policy Optimization (PPO).
- Enforcing chemical constraints such as atom valency, bond distances, and steric clash avoidance.
- Incorporating GSK3B, DRD2, and hERG bioactivity predictions into the reward function.
- Comparing the proposed model with PPO, DQN, heuristic, and random baseline approaches.
- Providing visualization and interactive interfaces for molecular generation and benchmarking.

1.6 Methodology Overview

This research implements a structured deep reinforcement learning pipeline for spatially aware molecular generation. The overall methodology consists of the following stages:

- **Protein and Pocket Preparation:**
  - Protein binding pocket structures are prepared for computational representation.
  - Relevant pocket regions are converted into a structured spatial representation.
- **Pocket Voxelization:**
  - Binding pockets are represented using a $16 \times 16 \times 16$ voxel grid.
  - Four channels encode hydrophobic, hydrogen-donor, electrostatic, and molecule occupancy information.
- **Environment Design:**
  - The BindingPocketEnv models molecule construction as a sequential decision-making process.
  - The agent selects atom types and three-dimensional coordinates at each step.
  - Chemical validity, valency, bond distances, and steric clashes are checked during generation.
- **Model Development:**
  - A 3D CNN actor-critic architecture is used for spatial feature extraction.
  - CBAM attention mechanisms identify important channel and spatial features.
  - Separate actor and critic heads generate actions and estimate state values.
- **Training and Validation:**
  - PPO is used to optimize the molecular generation policy.
  - Morgan fingerprints and Random Forest classifiers are used for bioactivity prediction.
  - The reward incorporates activity, safety, molecular quality, novelty, and generation efficiency.
- **Evaluation and Benchmarking:**

  - Generated molecules are evaluated for chemical validity, bioactivity, novelty, and structural quality.
  - The proposed 3D-PPO+CBAM model is compared with four baseline approaches.

1.7 Challenges and Limitations

During the development and implementation of the proposed molecular generation framework, several challenges were considered:

- **Chemical Validity:** Sequential atom placement can result in invalid valencies, unsuitable bond distances, or incomplete molecular structures. Chemical validation is therefore required throughout the generation process.
- **Spatial Representation:** Converting continuous protein binding pockets into voxel grids introduces discretization and resolution constraints. The selected voxel representation must retain meaningful spatial and chemical information while remaining computationally manageable.
- **Multi-Objective Optimization:** The agent must balance several competing objectives, including target activity, hERG safety, molecular novelty, chemical validity, and efficient molecule construction.
- **Computational Complexity:** Training a 3D CNN-based reinforcement learning model involves repeated environment interactions and volumetric convolution operations, resulting in significant computational requirements.
- **Generalization:** Molecular generation policies learned for one protein target may not directly transfer to another target with a different pocket geometry and chemical environment.

1.8 Significance of the Study

This project is significant because it demonstrates how deep reinforcement learning and three dimensional spatial representations can be combined to support automated molecular design. It is significant from both scientific and technological perspectives.

Scientific Significance:

- The project demonstrates the application of reinforcement learning to sequential molecular construction within 3D protein environments.
- The integration of CBAM attention enables the model to focus on chemically and spatially relevant regions of the binding pocket.
- The reward framework combines target activity, safety, novelty, and molecular quality within a single optimization process.
- The framework provides a basis for investigating target-specific molecular generation across multiple protein targets.

Technological Significance:

- The system integrates molecular generation, chemical validation, property prediction, visualization, and benchmarking into a unified framework.
- Interactive web and desktop interfaces make the generation process easier to visualize and analyze.
- The framework can support automated exploration of chemical space and reduce dependence on manually predefined molecular libraries.
- The approach provides a foundation for future drug discovery systems that combine deep reinforcement learning with three-dimensional protein structure information.

Chapter 2

Literature survey

2.1 Introduction

De novo drug design has increasingly adopted deep learning and generative models to explore large chemical spaces and generate molecules with desired properties. Early approaches mainly represented molecules as sequences or graphs, while recent methods increasingly incorporate three-dimensional molecular structure and protein binding-pocket information. Reinforcement learning has also been used to optimize generated molecules according to multiple objectives such as biological activity, drug-likeness, and molecular properties. These developments provide the foundation for target-aware molecular generation using both spatial information and reward-based optimization.

2.2 Previous Research

Cheng et al. (2021) reviewed the application of deep generative models to de novo molecular design and discussed different molecular representations and generation strategies. The study highlighted the potential of generative models for producing novel compounds with desirable properties while also identifying challenges related to molecular validity and optimization [1].

Liu et al. (2021) proposed DrugEx v2, a Pareto-based multi-objective reinforcement learning framework for de novo drug design. The method optimized multiple molecular objectives simultaneously and demonstrated the usefulness of reinforcement learning for balancing competing drug-design requirements [2].

Li et al. (2021) introduced DeepLigBuilder, a structure-based generative model that generates three-dimensional molecules within protein binding sites. The method demon-

strated the importance of incorporating the spatial environment of a target protein during molecular generation [3].

Xie et al. (2022) reviewed advances in three-dimensional deep generative models for de novo drug design. The study emphasized that 3D generation can better represent molecular geometry and protein–ligand interactions compared with approaches based only on two-dimensional molecular representations [4].

Wang et al. (2022) provided a comprehensive evaluation of different deep generative architectures for molecular design. Their analysis considered target-specific and goal-directed generation and showed that model performance depends strongly on the molecular representation and optimization objective [5].

Peng et al. (2022) proposed Pocket2Mol, an E(3)-equivariant generative model for molecular sampling conditioned on three-dimensional protein pockets. The model considers both spatial and chemical relationships within the pocket and demonstrated improved binding affinity and drug-related properties [6].

Atance et al. (2022) developed a reinforcement learning framework for fine-tuning graph-based molecular generators. The approach guided molecular generation toward desired properties, including predicted DRD2 activity, showing the effectiveness of reinforcement learning for goal-directed molecular optimization [7].

Wang et al. (2022) proposed RELATION, a three-dimensional generative model for structure-based de novo drug design. The framework incorporated protein– ligand geometric information and pharmacophore conditioning to generate molecules with favorable structural and binding characteristics [8].

Huang et al. (2022) introduced 3DLinker, an E(3)-equivariant variational autoencoder for generating molecular linkers. The method jointly considered molecular graphs and three-dimensional coordinates, demonstrating the importance of geometric information for constructing chemically feasible molecular structures [9].

Xu et al. (2022) proposed GeoDiff, a geometric diffusion model for molecular conformation generation. The method models atomic coordinates through an equivariant diffusion process and demonstrated improved generation of realistic molecular conformations [10].

Guan et al. (2023) introduced a 3D equivariant diffusion model for target-aware molecule generation and affinity prediction. The model jointly generates continuous

atom coordinates and categorical atom types while preserving three-dimensional geometric properties [11].

Zhang et al. (2023) proposed ResGen, a pocket-aware 3D molecular generation model based on parallel multiscale modelling. The framework explicitly models protein–ligand interactions and generates molecules within target pockets using a hierarchical generation process [12].

Wang et al. (2023) investigated molecular generation using A2C reinforcement learning. The study demonstrated how actor–critic reinforcement learning can be used to optimize generated molecules toward desired molecular properties, providing a relevant foundation for reinforcement learning-based molecular generation [13].

Du et al. (2023) presented a data-free reinforcement learning framework for structure-based de novo drug design. The method used molecular editing and search-based generation to optimize molecules according to structure-based objectives, addressing limitations associated with conventional atom-by-atom generation [14].

Svensson et al. (2024) systematically investigated reinforcement learning methods for de novo drug design, comparing different on-policy and off-policy approaches for molecular generation. The study demonstrated that reinforcement learning can effectively optimize molecular activity while maintaining structural diversity [15].

Jiang et al. (2024) proposed PocketFlow, a data-and-knowledge-driven structure-based molecular generative model. The framework incorporates chemical knowledge together with protein-pocket information to improve the generation of chemically valid and drug-like molecules [16].

Huang et al. (2024) introduced PMDM, a conditional diffusion model for three-dimensional molecule generation and lead optimization based on target pockets. The model uses local and global molecular dynamics to incorporate protein information and demonstrated its application to target-specific molecular generation and lead optimization [17].

2.3 Research gap

The reviewed studies demonstrate substantial progress in deep generative drug design, reinforcement learning, and three-dimensional target-aware molecular generation. How-

ever, many reinforcement learning approaches primarily operate on molecular sequences or graphs and therefore have limited direct representation of the spatial environment of a protein pocket. In contrast, recent 3D pocket-aware models provide improved geometric representation but are commonly based on autoregressive, variational, or diffusion-based generation rather than policy-based reinforcement learning.

A further limitation is that molecular validity, target activity, toxicity, novelty, and structural constraints are often handled using separate optimization or evaluation mechanisms. There is therefore a need for an integrated framework that represents the protein binding pocket in three dimensions, constructs the molecule sequentially, validates generated structures, and uses multiple molecular objectives within a reinforcement learning reward.

The proposed work addresses this gap by combining a voxelized 3D protein pocket representation with a 3D CNN and CBAM3D feature extractor, an actor–critic policy, and PPO-based optimization. The framework generates molecules through continuous three-dimensional coordinates and discrete atom types while incorporating binding activity, toxicity, novelty, structural validity, and molecular constraints into the reward formulation.

2.4 Summary of key insights

The literature indicates three major developments in automated molecular design. First, deep generative models have improved the ability to explore large chemical spaces and generate molecules with desired properties. Second, reinforcement learning provides an effective mechanism for goal-directed and multi-objective molecular optimization. Third, recent three-dimensional and protein-pocket-aware models demonstrate the importance of spatial information for structure-based drug design. These findings support the integration of three-dimensional pocket representation, reinforcement learning, and multi-objective reward optimization in the proposed framework.

2.5 Chapter Summary

This chapter reviewed recent research in deep generative molecular design, reinforcement learning, and three-dimensional protein-pocket-aware molecule generation. The reviewed studies demonstrated advances in multi-objective optimization, geometric molecular generation, and target-specific drug design. However, the integration of voxel-based 3D pocket representation, sequential molecular generation, and multi-objective PPO optimization remains less explored. This gap motivates the proposed reinforcement learning-based 3D molecular generation framework presented in the following chapters.

Chapter 3

Methodology

3.1 Introduction

The research approach used to develop a deep reinforcement learning-based framework for de novo molecular generation within three-dimensional protein binding pockets is described in this chapter. From protein structure preparation and molecular representation to environment design, model development, training, and evaluation, the methodology systematically explains each stage of the proposed framework.

The proposed method combines three-dimensional spatial representation with reinforcement learning to generate chemically valid molecules within protein binding pockets. The binding environment is represented as a structured voxel grid containing spatial and chemical information, while a Proximal Policy Optimization (PPO) agent learns to sequentially place atoms within the pocket. A three-dimensional Convolutional Neural Network (3D CNN) enhanced with a Convolutional Block Attention Module (CBAM) is used to extract relevant spatial features from the environment.

The generated molecules are evaluated using chemical validity constraints, target-specific bioactivity predictions, molecular quality, novelty, and safety-related properties. This methodological design enables the system to learn spatially aware molecular generation policies while providing a framework for automated and target-specific drug discovery.

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3.2 Problem Statement

3.2.1 Core Problem Definition

Traditional drug discovery involves searching through an extremely large chemical space to identify molecules that can effectively interact with a desired protein target. Conventional approaches such as virtual screening and manually guided molecular design can be computationally expensive and may require extensive expert intervention. Furthermore, generated molecules must satisfy multiple requirements simultaneously, including chemical validity, spatial compatibility with the protein binding pocket, target-specific activity, novelty, and acceptable safety characteristics.

Existing molecular generation approaches may primarily focus on molecular structure or chemical properties without explicitly considering the three-dimensional spatial environment of a protein binding pocket. Consequently, there is a need for an automated framework that can learn how to construct molecules directly within a spatially represented binding environment while receiving feedback from chemical and biological evaluation criteria.

Thus, the central problem addressed in this study is:

> ***"To design and develop a deep reinforcement learning-based system that generates chemically valid and spatially compatible molecules within three-dimensional protein binding pockets while optimizing target-specific bioactivity, molecular quality, novelty, and safety."***

3.2.2 Sub-Problems Identified

De novo molecular generation within protein binding pockets requires the simultaneous consideration of spatial relationships, chemical constraints, and biological properties. An inappropriate atom placement may result in invalid bonding, steric clashes, or chemically unrealistic structures. Similarly, optimizing only for target activity may result in molecules with poor safety or limited novelty. Therefore, an intelligent generation framework must incorporate multiple sources of feedback during the molecule construction process.

The following sub-issues were identified during this investigation:

1. Representing three-dimensional protein binding pockets as compact and informative voxel environments while preserving important spatial and chemical characteristics.

2. Sequential atom placement may produce chemically invalid molecules, requiring constraints for bond distances, valency, steric clashes, and hydrogen balance.

3. Simultaneously optimizing target-specific bioactivity, molecular quality, novelty, and safety makes the reward design and reinforcement learning process challenging.

4. Training a 3D deep reinforcement learning model is computationally demanding due to the large spatial state representation and repeated environment interactions.

5. A policy trained for one protein target may not directly generalize to different binding-pocket geometries, requiring evaluation and fine-tuning across multiple protein targets.

3.2.2 Data Sources

The proposed molecular generation framework requires two major categories of data: protein structural information for constructing three-dimensional binding-pocket environments and molecular bioactivity data for evaluating generated compounds. Protein structures are used to identify relevant binding-pocket regions and represent their spatial characteristics in the reinforcement learning environment.Bioactivity data for GSK3B, DRD2, and hERG are used to train Random Forest classifiers that estimate target-specific molecular activity.

3.2.3 Dataset Attributes

The protein binding-pocket environment is represented using a three-dimensional voxel grid of size $16 \times 16 \times 16$. Each voxel contains four channels representing different spatial and chemical properties of the binding environment. The molecular generation process uses six possible atom or termination actions:

- Carbon (C)
- Nitrogen (N)

- Oxygen (O)
- Fluorine (F)
- Chlorine (Cl)
- Stop (STOP)

For bioactivity prediction, Morgan fingerprints with radius 2 and 1024 bits are generated for each molecule. Random Forest classifiers containing 100 trees are trained separately for GSK3B, DRD2, and hERG activity prediction.

3.2.4 Data Partitioning

The bioactivity datasets are prepared for training the molecular property prediction models, while the reinforcement learning agent learns through repeated interactions with the molecular generation environment rather than through a conventional fixed training, validation, and testing split.

The Random Forest models are trained using the available curated bioactivity data and subsequently used to provide activity predictions during reinforcement learning. The reinforcement learning environment independently generates molecular trajectories through multiple episodes. Generated molecules are evaluated after construction using the learned bioactivity predictors and chemical validation procedures.

For generalization analysis, the trained framework is evaluated across multiple protein targets, including HSP90, SARS-CoV-2 Mpro, Abl kinase, EGFR, CDK2, and the μ-opioid receptor.

3.3 Data Preprocessing

In this step, protein structural information, molecular structures, and bioactivity information are prepared for use within the reinforcement learning framework. The main objectives are to construct consistent three-dimensional binding-pocket representations, generate molecular features for bioactivity prediction, and ensure that generated structures satisfy the required chemical constraints.

3.3.1 Validation and Cleaning of Data

Protein structures are processed to identify the relevant binding-pocket regions and remove information that is not required for the generation environment. The selected pocket regions are converted into the predefined voxel representation while maintaining their spatial relationships. These validation procedures ensure that the reinforcement learning agent receives meaningful feedback and that generated molecules can be reliably evaluated using downstream molecular property prediction models.

3.3.2 Normalization and Sequence Structure

The molecular generation environment represents the protein binding pocket using a normalized three-dimensional voxel grid of dimensions $16 \times 16 \times 16$ with four feature channels. This representation allows the 3D convolutional network to process spatial information consistently across different environment states.

At the beginning of each episode, a carbon atom is used as the initial molecular seed. The molecule is then constructed sequentially by selecting an atom type and its corresponding three-dimensional position.

The action at each generation step can be represented as:

$$a_t = (x_t, y_t, z_t, c_t)$$

where (x_t, y_t, z_t) represent the selected three-dimensional coordinates and c_t represents the selected atom type. The coordinates are constrained to the predefined voxel environment, while the atom type is selected from the available categorical actions.

3.3.3 Integration of Labels

Morgan fingerprints with radius 2 and 2048 bits are generated for each molecule and provided to the Random Forest classifiers for GSK3B, DRD2, and hERG activity prediction. The resulting activity predictions are incorporated into the reinforcement learning reward function to guide the agent toward molecules with desirable bioactivity and safety characteristics.

3.3.4 Standardization of Features

The protein binding-pocket environment is consistently represented using a $16 \times 16 \times 16$ voxel grid with four feature channels: hydrophobic, hydrogen-donor, electrostatic, and molecular occupancy. This fixed representation ensures that all environment states have the same input dimensions for the 3D CNN and CBAM-based feature extraction process.

3.4 Proposed Model Design

The proposed model combines three-dimensional convolutional feature extraction, attention-based feature refinement, and reinforcement learning to generate molecules within protein binding pockets. A 3D CNN extracts spatial information from the multi-channel voxel environment, while the CBAM mechanism emphasizes important spatial and channel-level features. The resulting representation is provided to an actor-critic network trained using Proximal Policy Optimization (PPO). The actor predicts the next molecular construction action, including the three-dimensional position and atom type, while the critic estimates the expected value of the current environment state.

3.4.1 Spatial Feature Extractor Based on 3D CNN

The 3D CNN component acts as the spatial encoder of the proposed framework. The input to the network is a four-channel voxel representation with dimensions $16 \times 16 \times 16$. Three convolutional blocks are used to progressively extract low-level and high-level spatial features. The convolutional layers use increasing channel dimensions of $4 \rightarrow 16 \rightarrow 32 \rightarrow 64$. Each convolutional block consists of a three-dimensional convolution operation followed by Batch Normalization and a Rectified Linear Unit (ReLU) activation. CBAM is subsequently applied to refine the extracted features.

The three-dimensional convolution operation can be represented as:

$$Y_{i,j,k} = \sum_{m,n,p} X_{i+m,j+n,k+p} \cdot K_{m,n,p}$$

where $\mathbf{X}$ represents the input feature volume, $\mathbf{K}$ represents the three-dimensional convolution kernel, and $\mathbf{Y}$ represents the resulting feature volume.Max-pooling operations are used to progressively reduce the spatial dimensions while retaining important features.

Finally, Global Average Pooling (GAP) is applied to produce a compact 256-dimensional spatial feature representation.

3.4.2 Attention-Based Feature Refinement Using CBAM

The Convolutional Block Attention Module (CBAM) is incorporated into the 3D CNN to improve the representation of important molecular and binding-pocket features. CBAM performs attention refinement through channel attention and spatial attention.
Channel attention allows the network to determine which feature channels contain the most relevant chemical or spatial information. Spatial attention subsequently identifies important regions within the three-dimensional feature representation.
The refined feature representation can be expressed as:

$$F' = M_s(M_c(F) \odot F)$$

where F represents the input feature map, M_c represents the channel attention map, M_s represents the spatial attention map, and $\odot$ denotes element-wise multiplication.This attention mechanism allows the model to focus on spatial regions and feature channels that are more relevant to molecular construction within the binding pocket.

3.4.3 Actor-Critic Output Layer

The final 128-dimensional feature vector obtained from the spatial encoder is provided to the actor and critic networks. For the continuous component, the actor predicts the mean and log-standard deviation for the three-dimensional coordinates (x, y, z). For the categorical component, a softmax distribution is used to select the next atom type from C, N, O, F, Cl, and STOP.
The critic produces a scalar value estimate representing the expected cumulative reward from the current state.
The actor action can therefore be represented as:

$$a_t = (x_t, y_t, z_t, c_t)$$

while the critic estimates:

$$V(s_t)$$

where s_t represents the current molecular and binding-pocket state.

Table 3.1: Configuration of the Proposed 3D-PPO Architecture

Layer Type	**Parameters**	**Output Dimension**	**Activation**
Input	$4 \times 16 \times 16 \times 16$	$4 \times 16 \times 16 \times 16$	–
3D CNN Block 1	16 channels	$16 \times 8 \times 8 \times 8$	ReLU
3D CNN Block 2	32 channels	$32 \times 4 \times 4 \times 4$	ReLU
3D CNN Block 3	64 channels	$64 \times 2 \times 2 \times 2$	ReLU
CBAM	Channel + Spatial Attention	$256 \times 2 \times 2 \times 2$	–
Global Average Pooling	–	128	–
Actor	XYZ + Atom Type	Action Distribution	–
Critic	Value Head	1	Linear

3.5 Reward Function and Model Training

The proposed reinforcement learning framework uses a chemistry-aware reward function to guide the agent toward the generation of molecules with desirable biological and structural characteristics. The reward combines bioactivity, molecular quality, novelty, and step-related components.

The total reward is defined as:

$$R_{\text{total}} = R_{\text{bioactivity}} + R_{\text{quality}} + R_{\text{novelty}} + R_{\text{step}}$$

The bioactivity component incorporates the predictions of the GSK3B, DRD2, and hERG classifiers.The GSK3B and DRD2 activity terms encourage desirable target activity, while the hERG term penalizes undesirable activity. Molecular novelty is evaluated using Morgan fingerprints and Tanimoto-based similarity, while molecular quality rewards valid structures and penalizes chemically invalid configurations.

3.5.1 Training Configuration

The reinforcement learning agent is trained using Proximal Policy Optimization (PPO). PPO is selected because it provides stable policy updates while limiting excessively large changes to the policy between optimization steps.

The primary training configuration is:

- Algorithm: Proximal Policy Optimization (PPO)
- Learning Rate: 3×10^{-4}
- Discount Factor (γ): 0.99
- GAE Parameter (λ): 0.95
- PPO Clipping Parameter (ϵ): 0.2
- Epochs per Batch: 4
- Mini-Batch Size: 16
- Entropy Coefficient: 0.01
- Maximum Atoms per Molecule: 15

During each episode, the agent begins with a carbon seed and sequentially selects atom positions and atom types. The environment evaluates each action according to chemical constraints and updates the molecular state. The episode terminates when the agent selects the STOP action or reaches the maximum permitted number of atoms.

The PPO clipped objective is expressed as:

$$L^{CLIP}(\theta) = E_t \left[\min \left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right]$$

where $r_t(\theta)$ is the probability ratio between the new and old policies, $\hat{A}_t$ is the estimated advantage, and ϵ is the clipping parameter.

The Generalized Advantage Estimation (GAE) mechanism is used to estimate the advantage values required for policy optimization. The entropy term is also incorporated to encourage sufficient exploration of the molecular action space.

3.5.2 Baseline Comparison and Evaluation

The proposed 3D-PPO+CBAM model is compared with multiple baseline approaches to evaluate the effectiveness of spatial attention and reinforcement learning. The baseline models include:

- 3D-PPO + CBAM
- 3D-PPO
- 3D-DQN
- Chemist Heuristic
- Random Generation

The generated molecules are evaluated using multiple chemical and reinforcement learning metrics. These include chemical validity, average episode reward, connected fragment rate, steric clash reduction, bioactivity scores, hERG safety, molecular weight, novelty, and convergence behaviour.
The proposed PPO+CBAM model achieved an average reward of +1.8 compared with +0.2 for the random baseline. The framework achieved a 100% chemical validity rate and a 98.5% connected fragment rate, while generated molecules generally contained between 8 and 15 atoms. The model also demonstrated improvements in target activity and safety-related evaluation measures.The proposed architecture was further evaluated across six protein targets to examine its generalization capability. The model retained approximately 65% of its reward after fine-tuning to new targets, demonstrating the ability to adapt its learned molecular generation policy to different binding-pocket environments.

3.5.3 Chapter Summary

This chapter presented the complete methodology used to develop the proposed deep reinforcement learning-based molecular generation framework. It described the problem formulation, sub-problems, protein and bioactivity data sources, dataset attributes, data preprocessing, molecular representation, and chemical validation procedures. The proposed 3D CNN–CBAM actor-critic architecture was also detailed, along with the chemistry-aware reward function incorporating bioactivity, molecular quality, novelty,

and safety. Finally, the training configuration, PPO optimization procedure, baseline models, and evaluation criteria were presented. The next chapter discusses the experimental results, comparative performance of the proposed model, molecular generation outcomes, and analysis of the obtained results.

Chapter 4

System design / Experimental setup

4.1 Introduction

The proposed system is designed for reinforcement learning-based three-dimensional de novo molecular generation within a protein binding pocket. The system formulates molecular construction as a sequential decision-making problem in which an agent learns to place atoms at suitable three-dimensional coordinates while considering the structural characteristics of the target binding pocket. The objective is to generate chemically valid and novel molecular structures that provide a suitable balance between multiple target-related properties and safety-related considerations.

The implementation uses a Proximal Policy Optimization (PPO) algorithm with an actor-critic architecture. The molecular environment represents the protein binding pocket as a three-dimensional voxel grid containing hydrophobic, hydrogen-donor, and electrostatic features, along with a channel representing the currently generated molecule. The agent receives this representation as its state and produces two components of an action: the three-dimensional coordinates at which an atom should be placed and the type of atom to be added. The available atom categories are carbon, nitrogen, oxygen, fluorine, chlorine, and a `STOP` token used to terminate molecule generation.

The system integrates deep learning, reinforcement learning, molecular structure processing, and reward-based optimization. A three-dimensional convolutional neural network with Convolutional Block Attention Modules (CBAM) extracts spatial features from the voxelized environment. The extracted representation is shared by the actor and critic components. The actor predicts both continuous atom coordinates and a categorical atom type, while the critic estimates the value of the current state. PPO then updates the policy using the rewards obtained from generated molecular structures.

4.2 System Design Overview

The system follows an agent-environment interaction cycle. Initially, the target protein binding pocket is converted into a three-dimensional voxel representation. Three channels represent pocket characteristics such as hydrophobic, hydrogen-donor, and electrostatic information, while a fourth channel records the occupancy of the molecule being constructed.

The agent receives this four-channel representation as its current state. It then selects an action consisting of a continuous three-dimensional coordinate and a discrete atom type. The supported atom categories are carbon, nitrogen, oxygen, fluorine, chlorine, and a `STOP` action. The `STOP` action terminates molecule construction and initiates final evaluation.

For every generated atom, the environment checks the resulting molecular structure. Invalid structures caused by conditions such as steric clashes or valency violations are terminated with a penalty. Valid actions update the molecular state and allow the agent to continue generation.

The generated molecule is evaluated using multiple objectives, including predicted GSK3B and DRD2 activity, hERG-related safety, molecular novelty, fragmentation, and molecular size. The resulting reward is used by the PPO algorithm to update the policy. The training process continues for multiple episodes and iterations while recording reward and loss values for monitoring.

The overall system flow can be summarized as:

Protein Structure → Binding Pocket Representation → Initial Molecular State → 3D Actor–Critic Network → Atom Position and Type Selection → Molecular Validation → Reward Calculation → PPO Update → Improved Molecular Generation

4.3 System Architecture

The proposed system architecture, shown in Fig. 4.1, consists of seven major stages: protein pocket representation, molecular generation environment, 3D feature extraction, actor–critic decision making, molecular validation, reward evaluation, and PPO optimization. The system progressively converts the target protein structure into a spatial state

representation, generates molecular actions, evaluates the resulting structure, and uses the obtained reward to update the policy.

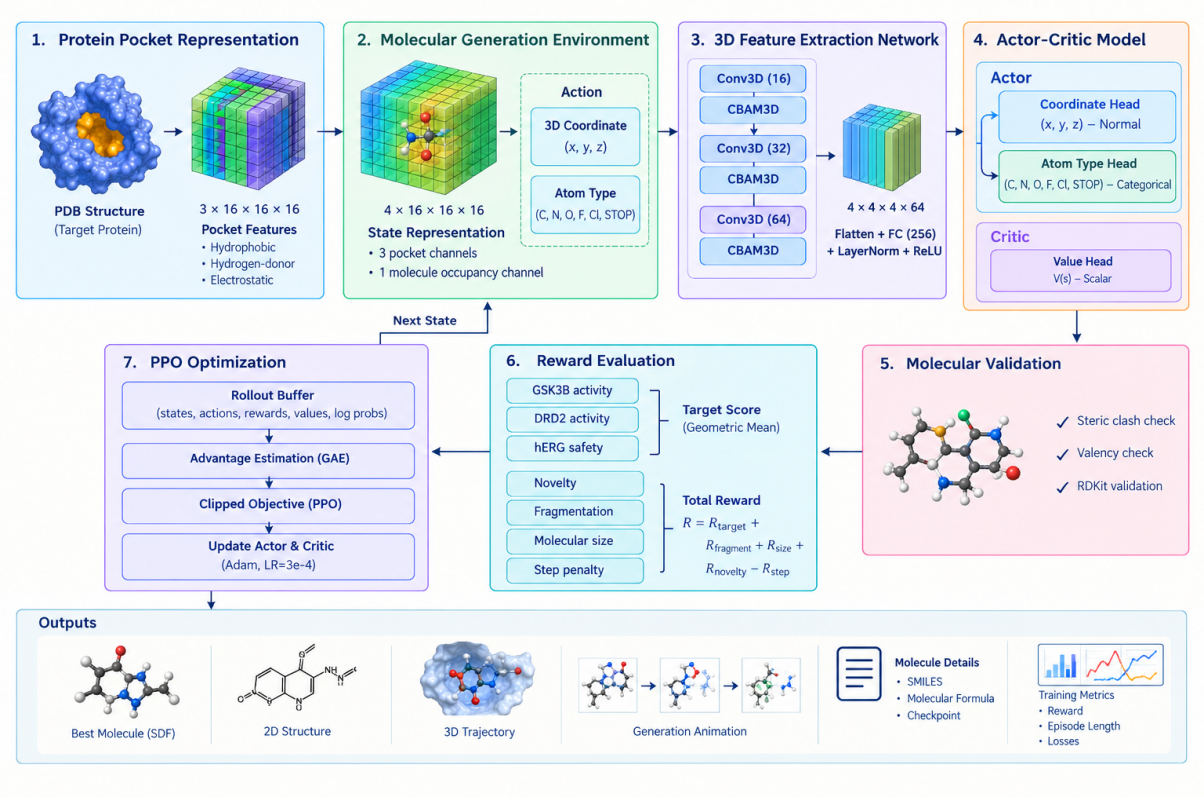


Figure 4.1: System architecture of the proposed reinforcement learning-based 3D molecular generation framework.

Protein Pocket Representation

The first stage represents the target protein binding pocket as a three-dimensional voxel grid. The protein structure obtained from the PDB is converted into three feature channels representing hydrophobicity, hydrogen-donor regions, and electrostatic characteristics. Each channel is represented on a $16 \times 16 \times 16$ grid. Therefore, the initial pocket representation is

$$P \in R^{3\times16\times16\times16}. \tag{4.1}$$

These channels provide spatial information about the binding pocket that can be used by the agent when deciding where to place new atoms.

Molecular Generation Environment

The molecular generation environment extends the three pocket channels with a fourth channel representing the occupancy of the molecule generated so far. Hence, the state provided to the neural network is

$$S_t \in R^{4\times16\times16\times16}. \tag{4.2}$$

The first three channels represent the protein pocket, while the fourth channel changes as atoms are added. The agent selects an action consisting of a three-dimensional coordinate and an atom type:

$$a_t = (x_t, y_t, z_t, e_t). \tag{4.3}$$

The available atom types are carbon (C), nitrogen (N), oxygen (O), fluorine (F), chlorine (Cl), and the STOP action. The STOP action terminates molecular construction.

3D Feature Extraction Network

The four-channel state is passed to the 3D feature extraction network. It contains three Conv3D layers with 16, 32, and 64 filters, respectively. The first convolution changes the input feature depth from 4 to 16, while the subsequent layers increase the feature representation to 32 and 64 channels. The second and third convolutional layers use a stride of 2, reducing the spatial dimensions from $16 \times 16 \times 16$ to $8 \times 8 \times 8$ and finally to $4 \times 4 \times 4$.

A CBAM3D (Convolutional Block Attention Module) follows each convolutional layer. It applies channel and spatial attention to emphasize informative regions of the protein pocket and generated molecular structure.

After the final convolution and attention operation, the feature representation has the form

$$F_t \in R^{64\times4\times4\times4}. \tag{4.4}$$

This representation is flattened and passed through a fully connected layer with 256 units, followed by Layer Normalization and ReLU activation. The resulting shared feature vector is used by both the actor and critic.

Actor–Critic Model

The shared 256-dimensional representation is provided to two branches: the actor and the critic.

The actor contains two output heads. The coordinate head predicts the mean of a Normal distribution for the three-dimensional atom position:

$$(x_t, y_t, z_t) \sim \mathcal{N}(\mu_t, \sigma_t). \tag{4.5}$$

The atom type head produces a categorical distribution over C, N, O, F, Cl, and STOP. Thus, the actor determines both the position and type of the next atom.

The critic contains a value head that produces a scalar state-value estimate:

$$V(S_t) = \text{Critic}(F_t). \tag{4.6}$$

This value estimates the expected future return from the current molecular state and is used during PPO optimization.

Molecular Validation

The selected action is passed to the molecular validation stage. The proposed atom is incorporated into the current molecular structure and checked using RDKit-based processing. The system performs steric clash checking, valency checking, and molecular validity checking.

If the generated structure violates the required constraints, the episode can be terminated with an appropriate penalty. If the action is valid, the molecular occupancy channel is updated and the resulting state is returned to the environment for the next generation step.

Reward Evaluation

For a completed molecule, the reward module evaluates multiple molecular properties, including GSK3B activity, DRD2 activity, hERG safety, novelty, fragmentation, molecular size, and step penalty.

The three target-related scores are combined using a geometric mean:

$$R_{\text{target}} = 10 \left(S_{\text{GSK3B}} S_{\text{DRD2}} S_{\text{hERG}} \right)^{1/3}. \tag{4.7}$$

The overall reward is then obtained by combining the target score with the additional molecular properties:

$$R = R_{\text{target}} + R_{\text{novelty}} + R_{\text{size}} + R_{\text{fragment}} - R_{\text{step}}. \tag{4.8}$$

This multi-objective reward encourages the generation of molecules that provide a balance between predicted activity, safety, structural quality, and novelty.

PPO Optimization

The generated states, actions, rewards, value estimates, and policy probabilities are stored in the rollout buffer. PPO uses these values to calculate the advantage and update the actor and critic.

The temporal-difference error used for advantage estimation is

$$\delta_t = r_t + \gamma V(S_{t+1})(1 - d_t) - V(S_t). \tag{4.9}$$

The PPO probability ratio is

$$r_t(\theta) = \frac{\pi_\theta(a_t|S_t)}{\pi_{\theta_{\text{old}}}(a_t|S_t)}. \tag{4.10}$$

The clipped objective is

$$L^{CLIP} = E_t \left[\min \left(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t \right) \right]. \tag{4.11}$$

The updated actor and critic are then used for subsequent molecular-generation episodes, creating the feedback loop shown in Fig. 4.1.

4.4 Experimental Setup

The system is implemented in Python using PyTorch for the neural network and reinforcement learning components and RDKit for molecular construction and validation. The molecular environment uses a $16 \times 16 \times 16$ voxel grid with four input channels, consisting of three protein-pocket features and one molecular occupancy channel.

The actor–critic model uses three Conv3D layers with 16, 32, and 64 channels, with CBAM3D attention modules for spatial and channel-wise feature refinement. The resulting representation is mapped to 256 shared features before being passed to the actor and critic branches.

PPO training uses the Adam optimizer with a learning rate of 3×10^{-4}. The discount factor is set to $\gamma = 0.99$, the GAE parameter to $\lambda = 0.95$, and the clipping parameter to $\epsilon = 0.2$. Each PPO update is performed for four epochs using a mini-batch size of 16. The value-loss and entropy coefficients are set to 0.5 and 0.01, respectively.

During training, the agent interacts with the environment over multiple episodes and stores the resulting states, actions, rewards, and value estimates for PPO updates. Training performance is monitored using average reward, episode length, policy loss, and value loss. The best generated molecule is saved along with its SDF structure, SMILES representation, molecular formula, 2D structure, and 3D generation trajectory.

Table 4.1: Experimental configuration of the proposed system

Parameter	**Configuration**
Learning algorithm	PPO
Network architecture	3D Actor–Critic
Grid size	$16 \times 16 \times 16$
Input channels	4
Feature extractor	3D CNN + CBAM3D
Convolutional channels	16, 32, 64
Shared feature dimension	256
Atom types	C, N, O, F, Cl
Termination action	STOP
Optimizer	Adam
Learning rate	3×10^{-4}
Discount factor	0.99
GAE λ	0.95
PPO clipping ϵ	0.2
PPO epochs	4
Mini-batch size	16

4.5 Chapter Summary

This chapter presented the system design and experimental workflow of the proposed reinforcement learning-based 3D molecular generation system. The design overview described the integration of protein pocket representation, molecular state construction, 3D feature extraction, and actor–critic decision making for sequential molecule generation. The system architecture explained the interaction between molecular validation, multi-objective reward evaluation, and PPO-based policy optimization. The experimental setup, including the model configuration, training parameters, and output generation process, was also outlined. Overall, this chapter establishes the implementation framework of the proposed system and provides the foundation for the performance analysis and results discussed in Chapter 5.

Chapter 5

Results and Discussion

5.1 Introduction

This chapter presents the results obtained from the proposed reinforcement learning-based framework for three-dimensional molecular generation. The evaluation focuses on the quality of the generated molecules, multi-objective reward performance, molecular validity, novelty, target-related properties, and comparison with alternative generation approaches. The generated molecule and system outputs are also examined to demonstrate the practical operation of the proposed framework.

5.2 Molecular Generation Results

The proposed framework successfully generated molecules within the selected three-dimensional protein binding pocket using the 3D-PPO with CBAM attention model. The generation process represents the protein pocket using spatial interaction information and sequentially places atoms according to the policy output. The generated molecule is evaluated using molecular validity and multiple target-related objectives before being considered as a candidate.

Figure 5.1 shows the molecular generation interface for the selected target pocket. The interface displays the protein pocket, the generated molecular structure, the number of placed atoms, the selected atom type, and the corresponding oracle and molecular properties. The system also provides policy diagnostics, including the probability distribution over possible atom types and the critic value.

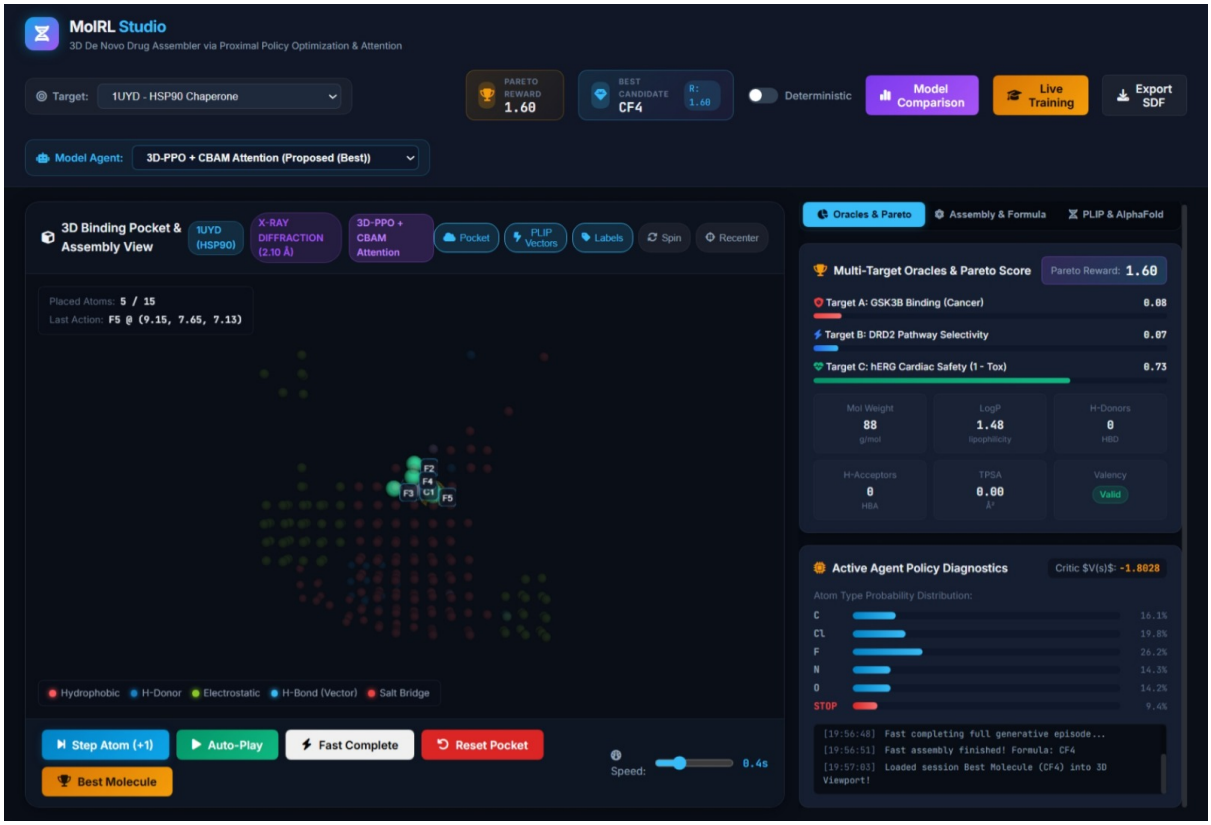


Figure 5.1: Molecular generation and evaluation interface of the proposed 3D-PPO with CBAM attention framework.

For the shown generation, the system completed the molecular assembly with five placed atoms and produced a valid candidate with the molecular formula CF_4. The generated molecule obtained a Pareto reward of 1.60, with GSK3B, DRD2, and hERG scores of 0.08, 0.07, and 0.73, respectively. The molecular properties displayed by the system also indicate a molecular weight of 88 g/mol, a LogP value of 1.48, and valid valency.

5.3 Multi-Objective Performance and Model Comparison

The proposed 3D-PPO with CBAM attention model was evaluated against four alternative approaches: standard 3D-PPO without CBAM, random exploration, a chemist rule-based policy, and a 3D-DQN model. The comparison considered molecular validity, single-fragment generation, Pareto reward, GSK3B affinity, hERG safety, novelty, and

inference latency.

Table 5.1: Comparison of molecular generation approaches

Model	Valid.	1-Frag.	Pareto	GSK3B	hERG	Novel	Latency
Standard 3D-PPO	100%	100%	7.14	0.160	0.684	100%	4.1 ms
Random Exploration	100%	100%	6.31	0.127	0.637	100%	1.1 ms
3D-PPO + CBAM	100%	100%	5.35	0.080	0.726	100%	9.4 ms
Chemist Rule-Based	100%	100%	5.19	0.040	0.736	100%	0.6 ms
3D-DQN	100%	100%	5.00	0.010	0.756	100%	1.9 ms

Figure 5.2 presents the corresponding benchmarking dashboard and radar visualization for the five evaluated approaches. All models achieved 100% validity and 100% single-fragment generation in the reported evaluation. The proposed model achieved a Pareto reward of 5.35, with a GSK3B score of 0.080 and an hERG safety score of 0.726. It also maintained 100% novelty while requiring 9.4 ms latency.

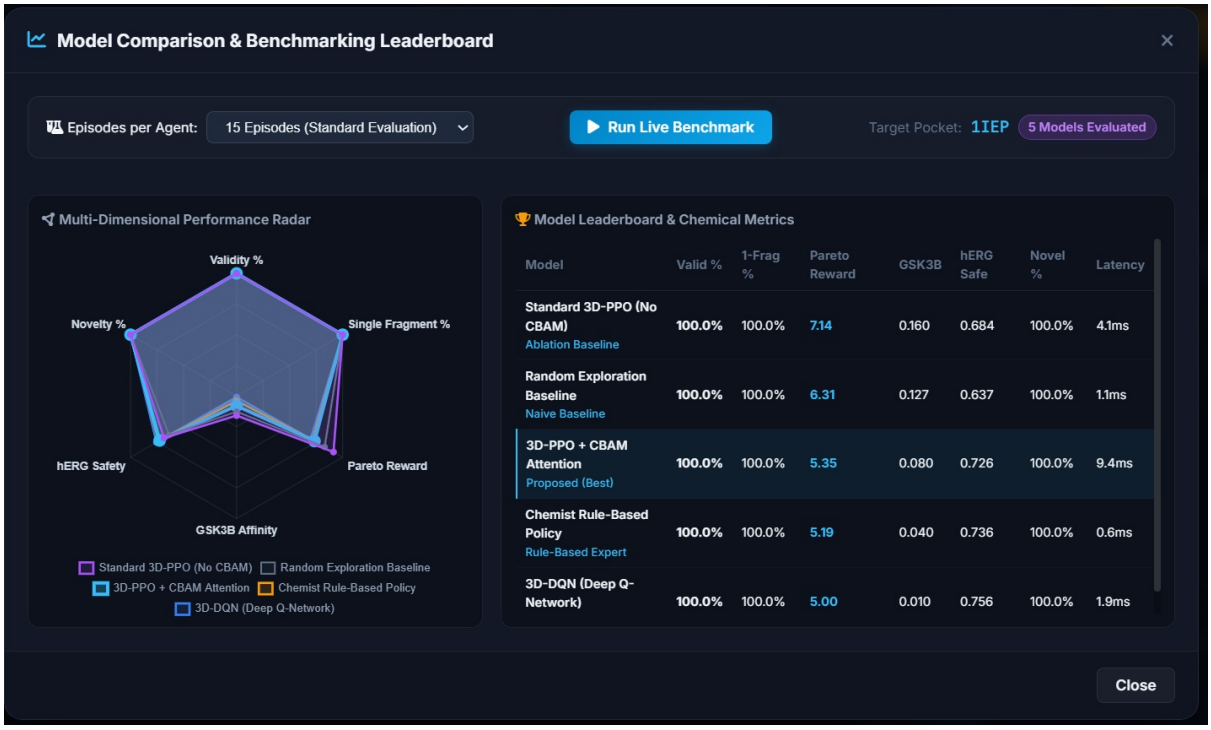


Figure 5.2: Model comparison and benchmarking results for the evaluated molecular generation approaches.

The comparison shows that the proposed model does not obtain the highest Pareto reward across all evaluated approaches. However, it provides a balanced multi-objective

performance, achieving a GSK3B score of 0.080, hERG safety of 0.726, and 100% validity, single-fragment generation, and novelty. The results indicate that CBAM-based feature attention provides a different trade-off between target-related objectives and molecular properties compared with the other evaluated approaches.

5.4 Best Generated Molecule and Output Analysis

The session-level best candidate generated by the proposed framework is shown in Figure 5.3. The molecule has the formula CF_4 and the SMILES representation FC(F)(F)F. It obtained a Pareto reward of 1.60 for the displayed generation session. The corresponding target scores were 0.08 for GSK3B, 0.07 for DRD2, and 0.73 for hERG safety.

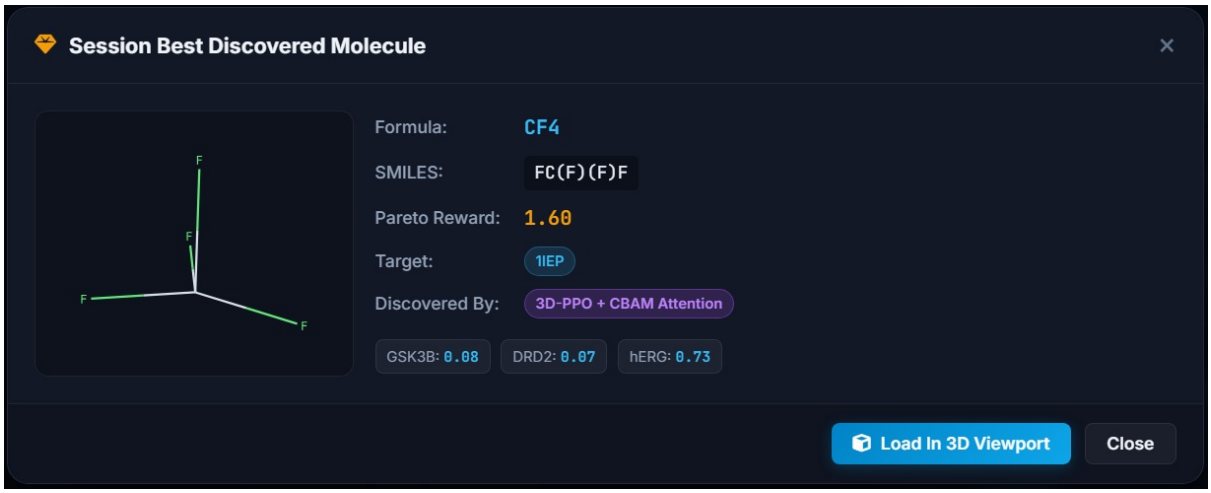


Figure 5.3: Session best molecule generated by the proposed framework.

The result demonstrates that the framework can generate a chemically valid candidate and report its molecular representation together with the multi-objective oracle scores. The generated structure can subsequently be loaded into the three-dimensional molecular viewer for further inspection.

5.5 Overall Results Discussion

The experimental results demonstrate that the proposed framework successfully performs sequential three-dimensional molecular generation within a protein binding pocket. Across the benchmarked approaches, all evaluated models achieved 100% molecular va-

lidity and 100% single-fragment generation in the reported evaluation. The proposed 3D-PPO with CBAM attention achieved 100% novelty and an hERG safety score of 0.726, while obtaining a GSK3B score of 0.080.

The comparison also shows that different generation strategies provide different trade-offs between target-related objectives, safety, reward, and computational latency. The proposed model therefore demonstrates the feasibility of combining three-dimensional pocket information, attention mechanisms, and PPO-based molecular generation for multi-objective drug design.

5.6 Summary

This chapter presented the experimental results of the proposed reinforcement learning-based 3D molecular generation framework. The generated molecule, molecular properties, multi-objective scores, and model benchmarking results were analyzed. The proposed 3D-PPO with CBAM attention successfully generated valid and novel molecular candidates and demonstrated a balanced performance across the evaluated objectives. The results provide evidence of the practical feasibility of the proposed framework for three-dimensional target-aware molecular generation.

Chapter 6

Conclusion

6.1 Summary of the Project

The proposed project developed a reinforcement learning-based framework for three-dimensional de novo molecular generation within a protein binding pocket. The system represents the binding pocket as a voxelized 3D environment and uses a 3D convolutional neural network with CBAM attention to extract spatial features. An actor–critic architecture with Proximal Policy Optimization (PPO) was then used to sequentially generate molecular structures by selecting atom types and their three-dimensional positions.

The generated molecules were evaluated using molecular validity and multiple objectives, including GSK3B activity, DRD2 activity, hERG safety, novelty, fragmentation, and molecular constraints. The framework also provides molecular representations such as SMILES, molecular formula, and three-dimensional visualization, enabling the generated candidates to be examined after the generation process.

6.2 Achievement of Objectives

The main objective of developing a three-dimensional reinforcement learning framework for target-aware molecular generation was achieved. The system successfully integrates protein-pocket representation, 3D feature extraction, attention-based feature refinement, actor–critic decision making, molecular validation, and PPO optimization into a single workflow.

The objective of incorporating multiple molecular properties into the generation process was also achieved through the multi-objective reward formulation. The experimental evaluation showed that the proposed 3D-PPO with CBAM attention generated valid and novel molecular candidates, with the benchmark reporting 100% molecular validity, 100%

single-fragment generation, and 100% novelty for the evaluated model. The proposed model also achieved a reported GSK3B score of 0.080 and hERG safety score of 0.726 in the benchmark evaluation.

6.3 Practical Implications

The developed framework demonstrates the practical potential of combining reinforcement learning and three-dimensional protein-pocket information for automated molecular design. Instead of manually constructing candidate molecules, the system can explore molecular structures sequentially and evaluate them according to multiple objectives.

The framework can therefore serve as a computational tool for early-stage drug discovery research by assisting in the exploration of candidate molecules. Its visualization and benchmarking components also provide a convenient way to inspect generated structures and compare different generation strategies.

6.4 Limitations

Despite the obtained results, the proposed framework has several limitations. The quality of generated molecules depends on the accuracy and coverage of the protein-pocket representation and the molecular evaluation functions used by the reward system. The oracle-based target scores are computational estimates and should not be interpreted as experimental evidence of biological activity or safety.

The generated candidates may also require additional chemical and pharmacological validation before being considered for practical drug development. Furthermore, the benchmark results are based on the evaluated target pocket and a limited set of generation approaches; therefore, the observed performance may not directly generalize to all protein targets or molecular design tasks.

6.5 Future Scope

The framework can be extended by evaluating a larger number of protein targets and incorporating more diverse molecular objectives. Future work can also investigate improved

reward functions that combine additional drug-likeness, synthetic accessibility, binding affinity, and toxicity criteria.

Further improvements can be made to the molecular generation policy by exploring alternative reinforcement learning and generative approaches, including improved PPO strategies and diffusion-based models. Experimental validation of generated candidates through molecular docking, molecular dynamics simulations, and ultimately laboratory-based testing would also provide stronger evidence of their practical usefulness.

The framework can additionally be optimized for faster generation and interactive visualization, allowing it to be used as a more efficient computational platform for structure-based de novo drug design.

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